Mobile Robots Exam – 2019-2010 – Duration: 2 h

Information about Multiple Choice Questions.

Multiple Choice Questions have a penalty system.

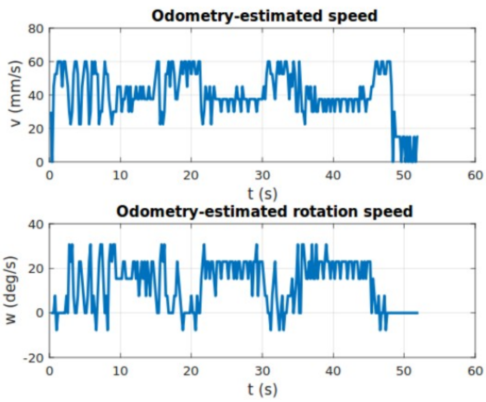
- Correct answer: 100%
- No answer: 0%
- Incorrect answer: -20% to -50% depending on the severity of the error.
- The idea is that ticking all answers as correct will usually result in a negative grade.

How to answer:

- Do not reproduce the question text or graphic.
- Just give the correct answer(s) by putting the question number and the list of correct answers, or “none” if none of the answers is correct. Examples:
  - Q25: True
  - Q26: B, D
  - Q27: None
  - Q28:
- The last example is what to do when you do not wish to answer the question. Beware of the difference between answering “none” and not answering!
- Any **clear** solution is OK. Several answers on one line to limit number of pages is possible:
  Q25: True   Q26: B, D   Q27: None

Exercise 1: Localization lab related questions (15 min).

Q1: As shown by the figure below, the estimated speed and rotation speed of the robot in the lab is noisy. What is the origin of this noise?



A) The operator shivers.
B) Bad low level control of the robot.
C) Quantization noise in the measurement of the angular position of the wheels.
D) Bad joystick.
E) Incorrect robot geometric parameters.

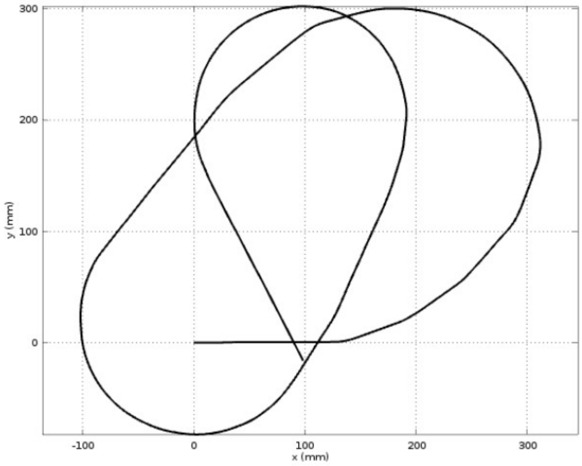
Q2: In the lab, odometry is performed using:

A) Wheel encoders only.
B) Wheel encoders and a gyrometer.
C) A gyrometer and a Doppler radar.
D) Magnets in the ground.
E) Magnets in the wheels and in the ground.

Q3: In the lab software, odometry equations are:

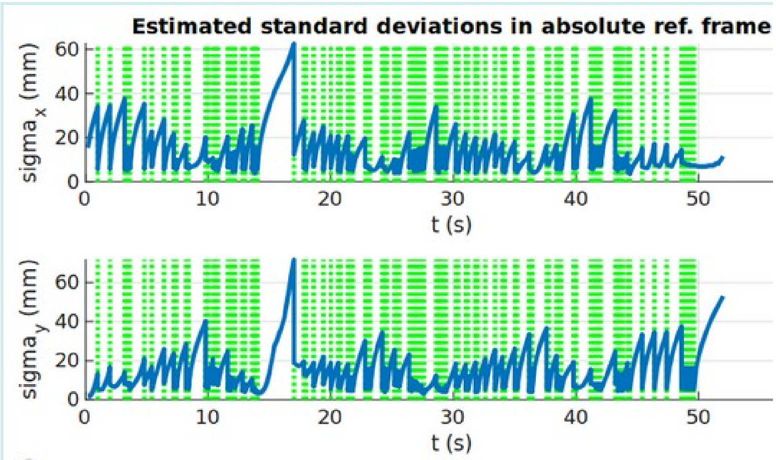
A) In the measurement equation
B) In the prediction step.
C) Neither in the measurement equation, nor in the prediction step.

Q4: Consider the graph below, which is an estimation of the robot path for data set TwoLoops.



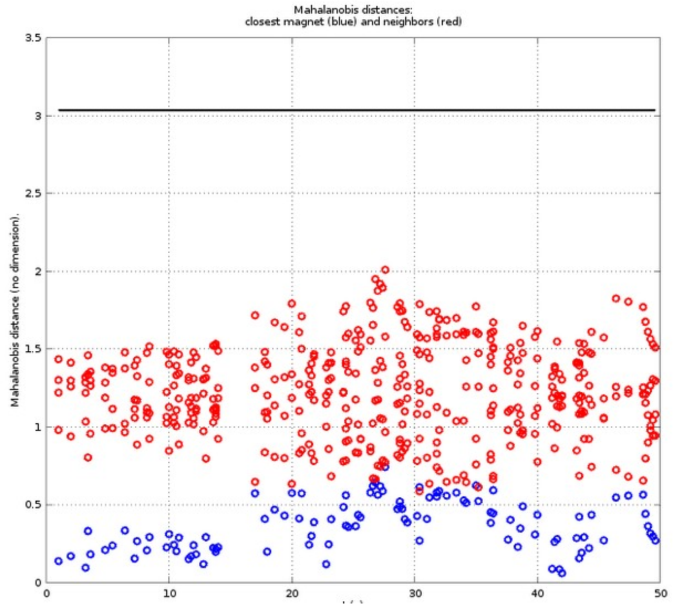
A) This path can be the result of an odometry-only path estimation.
B) An incorrectly tuned Kalman filter may give this result.
C) This path has been estimated by a correctly tuned Kalman filter.

Q5: Consider the graph below of the estimatead standard deviations of the state error.



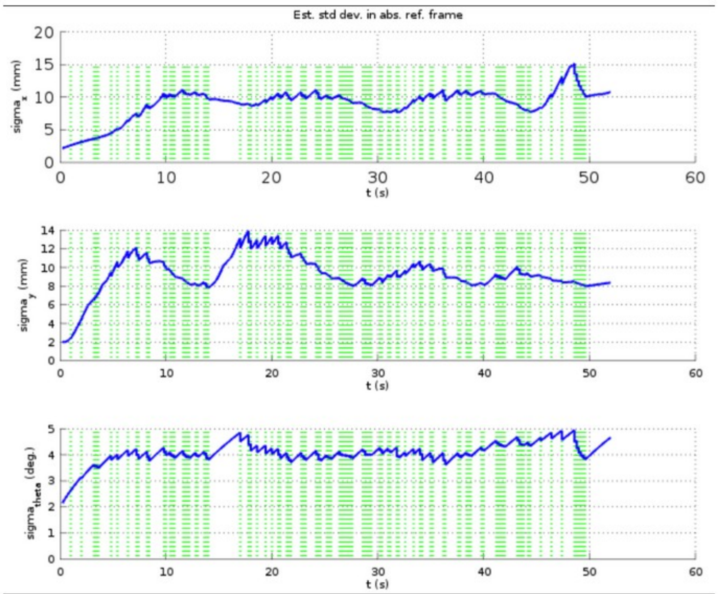
A) Apart from a single case, sigma_x and sigma_y are less than the distance between two magnets (55 mm), so these results are okay.
B) The estimated standard deviations are too high.
C) The estimated standard deviations are too low.

Q6: Consider the graph of the Mahalanobis distances below. Which parameter is incorrectly set?



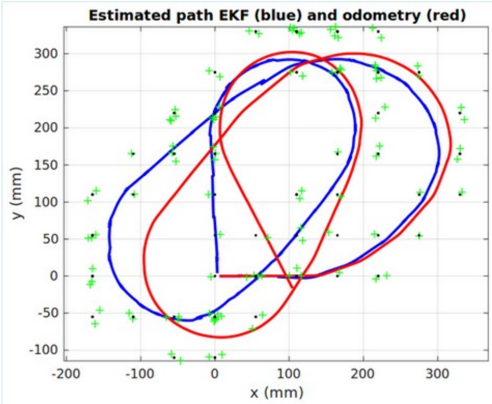
A) Pinit.
B) Mahalanobis distance threshold.
C) Qalpha (state noise)
D) Qgamma (measurement noise)
E) Qbeta (input noise) through incorrect sigmaTuning value.

Q7: With a correctly tuned EKF, the position uncertainties in the lab go down to the 1-2 mm range. It is not the case in the figure below.



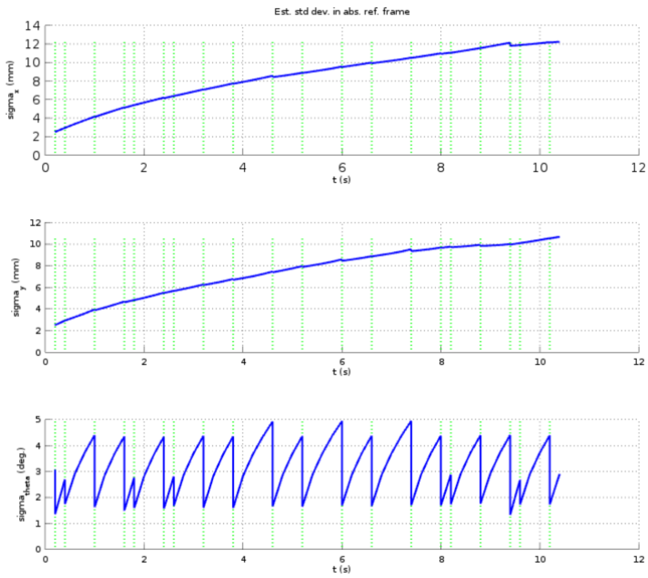
A) Sigma_tuning is too low. So the filter puts too much trust in its predicted state and does not let the update step modify it.
B) Qgamma has not been correctly determined. It is too high and thus, measurements do not effectively decrease the uncertainty.
C) Sigma_tuning is too high. It causes uncertainties to be higher.
D) Pinit has been set too high, and the estimator does not recover from this situation.

Q8: Among the various graphs that the lab software plot is the estimated path. By considering the graph below **only**, what can you say about the tuning of the EKF?



A) I have no indication that the tuning is incorrect, but I do not have enough information to assure that it is.
B) The position estimation returns to (0,0), and we know it was the case for the real path, so the EKF is correctly tuned.
C) The estimated path does not match odometry: the EKF is not correctly tuned.

Q9: You are experimenting with a localization system and you get the standard deviations given below.



Say whether you are satisfied of these results and give a very short justification (maximum two lines of text).

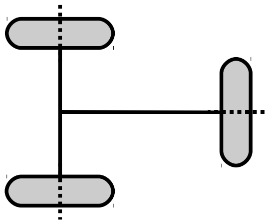
Q10: You have to develop a localization system based on a state estimator, *e.g.* an Extended Kalman Filter. Say whether you study observability first and then process and analyze real and/or simulated data, or whether you prefer to work the other way around, *i.e.* study observability after having processed and analyzed data. Justify you answer in a few lines of text (No more than 10).

Exercise 2 (30 min): Kinematics related short questions.

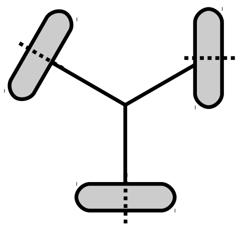
Question numbers continue in sequence with respect to exercise 1 on purpose.

For all justifications, clarity and conciseness will be important grading criteria.

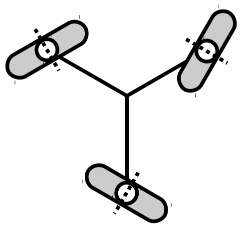
Q11: What is the degree of mobility of the platform equipped with the following configuration of wheels? Justify your answer in a maximum of three lines of text.



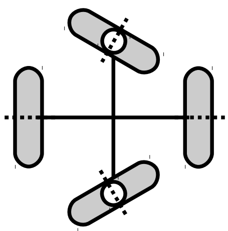
Q12: Same question with this wheel configuration. Justify your answer in a maximum of three lines of text.



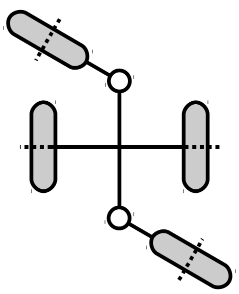
Q13: What is the type of this robot? Give a short justification (maximum five lines).



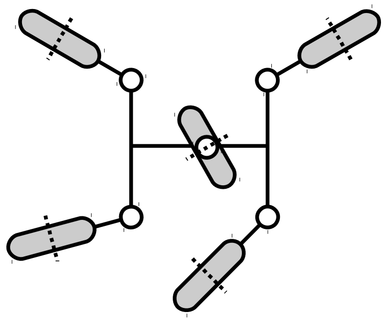
Q14: What is the type of this robot? Give a short justification (maximum five lines).



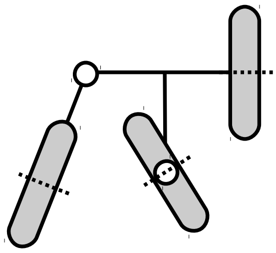
Q15: What is the type of this robot? No justification required.



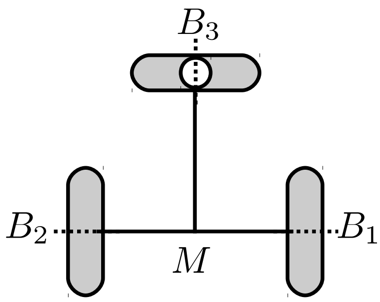
Q16: What is the type of this robot? No justification required.



Q17: What is the type of this robot? No justification required.

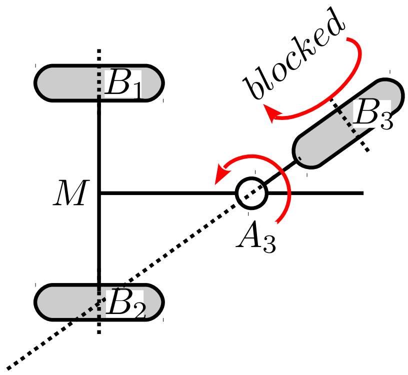


Q18: What is the location of the Instantaneous Center of Rotation?



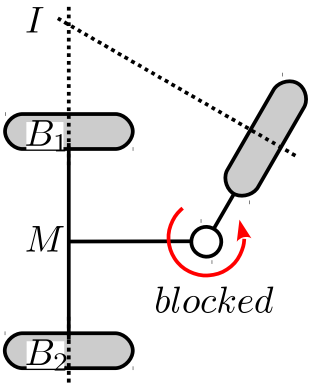
A) At B1
B) At M
C) At B3
D) At infinity
E) At B2

Q19: The spin angle of wheel 3 is blocked at its current position. What is the location of the Instantaneous Center of Rotation?



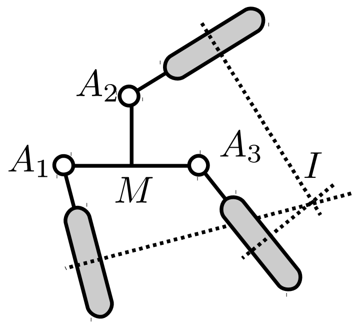
A) At M
B) At B1
C) At B2
D) At B3
E) At infinity on the axle of the fixed wheels.

Q20: The orientation angle of wheel 3 is blocked at its current position. What is the location of the Instantaneous Center of Rotation?



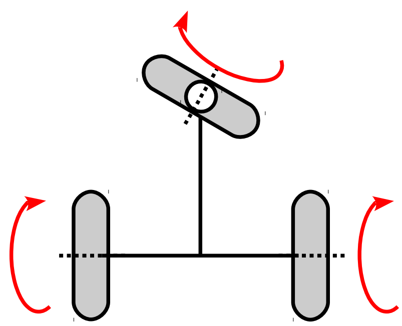
A) At B1
B) At B2
C) At I
D) At infinity
E) At M
F) Not enough information to tell.

Q21: What is the location of the Instantaneous Center of Rotation?



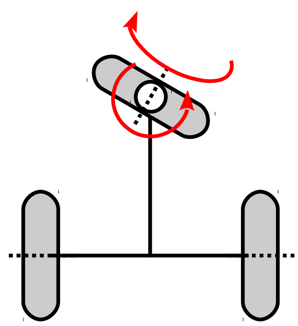
A) At A1
B) At A2
C) At A3
D) At infinity
E) At I
F) Not enough information to tell.

Q22: The figure shows **all** the motors of the robot. What can you say about this motorization?



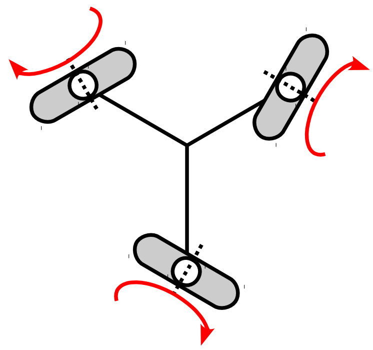
A) It is the standard motorization for this robot.
A) It is redundant but usable.
B) It is not usable.

Q23: The figure shows **all** the motors of the robot. What can you say about this motorization?



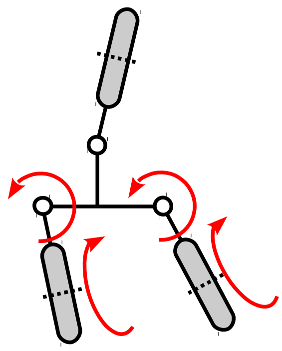
A) It is redundant but usable.
B) It will not allow to control the robot.
C) It is the standard motorization for this robot.

Q24: The figure shows **all** the motors of the robot. What can you say about this motorization?



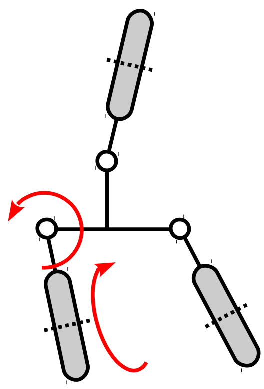
A) It is redundant but can be used.
B) It will not allow to control the robot.
C) It is the standard motorization for this robot.

Q25: The figure shows **all** the motors of the robot. What can you say about this motorization?



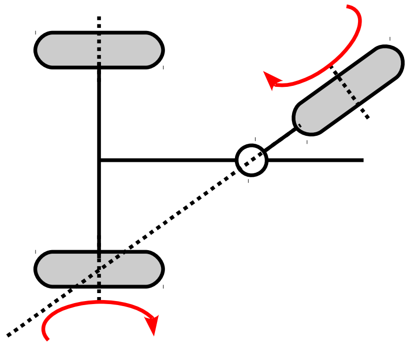
A) I can remove one of the motors (any one) and still have a perfectly functional robot.
B) It will not allow to control the robot.
C) It is a redundant motorization, necessary because of actuation singularities.
D) It is the standard motorization for this robot.

Q26: The figure shows **all** the motors of the robot. What can you say about this motorization?

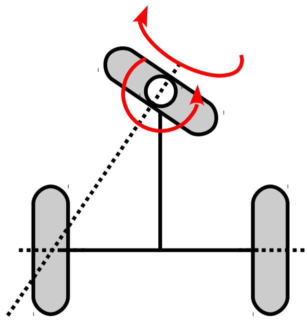


A) It is correct. I can orient the actuated wheel and move in the corresponding direction, so I can control the robot.
B) The robot is under-actuated. But I can control the x and y position along any path, just not the heading.
C) This motorization is incorrect, the robot is unusable.

Q27: All actuators of the robot are shown. State whether this configuration is an actuator singularity and shortly explain why (maximum five lines of text).



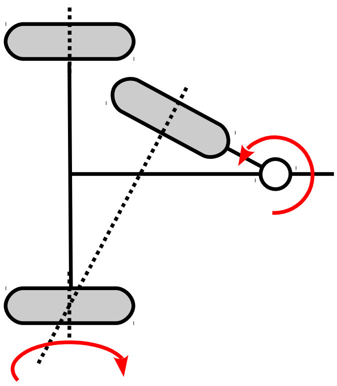
Q28: All actuators of the robot are shown. This configuration is an actuation singularity.



A) True

B) False

Q29: All actuators of the robot are shown. State whether this configuration is an actuator singularity and shortly explain why (maximum five lines of text).



Exercise 3 (15 min): Measurement equation

A robot is equipped with a GPS sensor. The sensor provides the position of its antenna G with respect to frame R_o (see figure 3.1). The position (a,b) of the GPS antenna with respect to the robot frame R_m is assumed known.

1. Assuming the state vector is the posture of the robot with respect to R_o, as depicted in figure 3.1, write a valid measurement equation for this system.
2. Write the matrix C that you would use in an EKF-based localization system for this robot and this measurement equation.

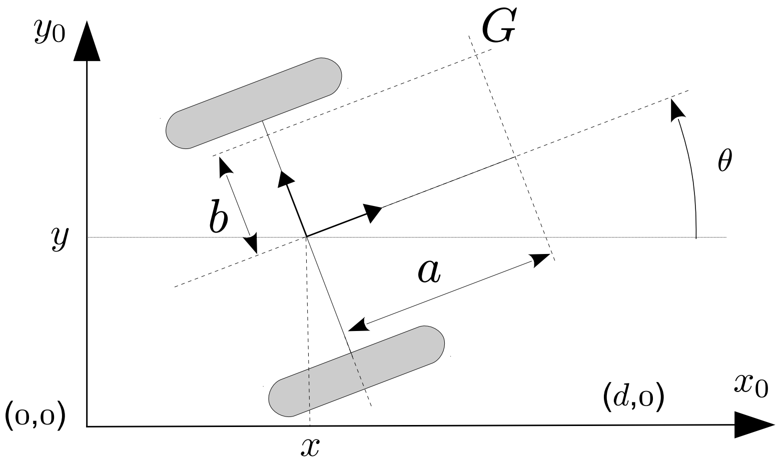


Figure 3.1. The robot and its GPS sensor antenna at point G.

Exercise 4 (20 min): Subset of an observability study.

We are going to partially study the observability of the posture of the robot of figure 3.1. The posture will be the state vector.

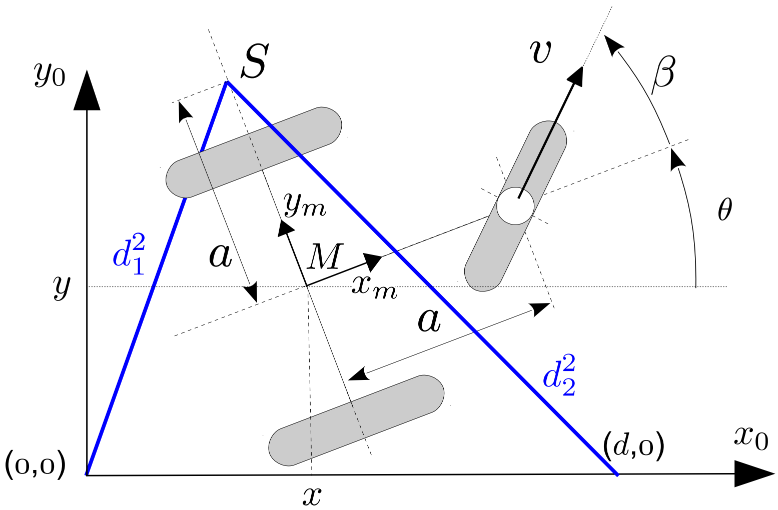


Figure 3.1.

The kinematics of the robot are given by the following set of equations:

$$\begin{cases} \dot{x} = v\,\cos\beta\,\cos\theta \\ \dot{y} = v\,\cos\beta\,\sin\theta \\ \dot{\theta} = (v/a)\,\sin\beta \end{cases}$$

The robot is equipped with two sensors which measure the **squared** distance from the sensor location S to the origin of the absolute reference frame and to point (d,0) of the absolute reference frame.

1. Establish the measurement equation g for this problem.

2. Calculate the gradients of the components of g.

3. Calculate the first order Lie derivates of the components of g.

4. Consider the sub-matrix of the observability matrix $\begin{vmatrix} dg_1 & dg_2 & dL_f g_1 \end{vmatrix}$ and, without calculating its determinant, find a case other than v=0, where the determinant of the matrix is 0. In that case, is the sufficient rank condition satisfied?

5. Again without calculating the determinant, is the problem solved by considering the sub-matrix $\begin{vmatrix} dg_1 & dg_2 & dL_f g_2 \end{vmatrix}$?

6. Do you think that the situation you discovered in questions 4 and 5 is a real observability problem? Explain how you could have obtained the same result without calculation.

Exercise 5 (20 mins): Robot modelling

W	L	α	d	β	γ	φ	ψ
$1f$	a	$-\pi/2$	0	0	$-\pi/2$	φ_{1f}	$-\pi$
$2f$	a	$\pi/2$	0	0	$\pi/2$	φ_{2f}	π
$3s$	b	0	0	β_{3s}	0	φ_{3s}	β_{3s}

Table 2.1. Parameters of the robot

1. Give the schematic representation of the robot. The figure should show frame R_m, vectors x_1, x_2 and x_3, the dimensions a and b, and the angle β_{3s}. Make sure β_{3s} is indicated by a circular arrow with the origin at the zero position of the angle. I recommend you first draw on scrap paper and copy a **clean result** on the answer sheet once you are sure of your result. Clarity and cleanliness of the figure will be evaluation criteria.

2. Calculate matrices J_1, C_1 and C_1*.

3. Calculate the degree of mobility and give the robot type.

4. Give a kernel of C_1*.

5. Give the posture kinematic model of the robot.

Exercise 6 (20 min): Motorization study.

We consider in this exercise the robot of figure 4.1. The robot has in practice at least on additional castor wheel, which is not motorized and is **completely ignored** in this exercise. Parameter r is the radius of both wheels. The configuration kinematic model of the robot is:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\delta}_1 \\ \dot{\delta}_2 \\ \dot{\varphi}_{1s} \\ \dot{\varphi}_{2s} \end{bmatrix} = \begin{pmatrix} 2\cos\delta_1\ \cos\delta_2\ \cos\theta - \sin(\delta_1+\delta_2)\ \sin\theta & 0 & 0 \\ \sin(\delta_1+\delta_2)\ \cos\theta + 2\cos\delta_1\ \cos\delta_2\ \sin\theta & 0 & 0 \\ \sin(\delta_1-\delta_2)\ /\ a & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2\cos\delta_2\ /\ r & 0 & 0 \\ 2\cos\delta_1\ /\ r & 0 & 0 \end{pmatrix} * \begin{bmatrix} u_m \\ u_{s1} \\ u_{s2} \end{bmatrix}$$

Study the motorization of this robot. You are expected to take into account the singularities, explain the physical meaning of the singularities and propose a motorization solution that is usable in practice, assuming there are no angular limits on angles δ_1 and δ_2.

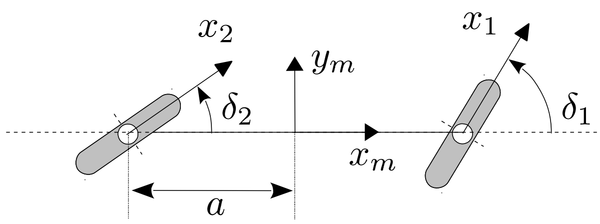


Figure 5.1. The robot to study.